

TAPAS KUMAR

Senior Python Backend Engineer | AI-Enabled Product Engineer

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PROFESSIONAL SUMMARY

Senior Python Backend Engineer with 5+ years of experience building production-grade backend systems, APIs, and AI-integrated applications. Experienced in designing scalable services, integrating LLM APIs into products, and ensuring system reliability through testing and validation. Strong focus on shipping clean, maintainable software and building real-world product features using Python.

CORE EXPERTISE

- Python Backend & Product Engineering: Python, FastAPI, REST APIs, backend system design, microservices, API integration, CLI applications, Git, Docker, Linux basics
- AI Integration (Practical Usage) : LLM API integration (GPT, LLaMA), prompt engineering, AI-powered feature development, basic RAG pipelines for product use cases
- Frontend & Full-Stack Awareness: React.js (basic to working knowledge), UI integration with APIs, full-stack application flow understanding
- Testing & System Reliability: API testing, integration testing, regression testing, debugging, system validation, production readiness
- Data Handling (Basic Support Role): SQL, data validation, ETL basics, business logic implementation

WORK EXPERIENCE

Simplyfi Softech India Pvt. Ltd.

Sep 2022 – Jan 2026

Applied AI Engineer — LLM Systems & Generative AI

Bengaluru, India

- Developed backend services and product features using Python and FastAPI, supporting production deployment, debugging, and system reliability improvements for enterprise applications.
- Integrated LLM-based capabilities into backend systems for document processing and automation workflows, improving feature efficiency and output quality.
- Worked with product and business teams to translate requirements into APIs, backend services, and workflow automations.
- Built CI/CD pipelines for deployment and monitoring of backend and AI-integrated services in production environments.
- Developed automation frameworks for API testing, integration testing, and system validation to ensure reliability of backend services and AI features.
- Built invoice processing system using LLMs for structured data extraction with REST API integration.
- Developed document processing system integrating OCR and object detection models for information extraction.
- Built automated reporting system generating structured outputs in PDF, Excel, and Word formats through backend workflows.

Think Dignity International

May 2021 – Aug 2022

AI/ML Engineer |

Remote, India

- Developed and deployed computer vision pipelines for real-time analytics systems in production environments.
- Integrated object detection models into video processing workflows for large-scale analytics applications.
- Performed system validation, performance testing, and reliability checks for deployed AI systems.
- Built real-time traffic analytics pipeline using video-based data processing and monitoring systems.
- Developed vehicle detection and tracking system for large-scale traffic monitoring use cases.
- Built speed estimation system using camera-based video analysis for real-time analytics applications

Innodatatics

Feb 2020 – May 2021

Data Engineer

Hyderabad, India

- Built ML-backed APIs and predictive analytics applications using Python and Flask for business intelligence use cases.
- Developed REST APIs for serving machine learning model predictions in production systems.
- Validated model outputs against business rules and test datasets to ensure correctness and reliability.

- Built employee attrition prediction system integrating ML models with backend APIs for real-time inference and response generation.

CERTIFICATIONS

- CS50x: Introduction to Computer Science — Harvard University
- Python 101 for Data Science (IBM) & Data Analysis with Python
- Fundamentals of AI Agents — by Hugging Face
- LangChain for LLM Application Development — DeepLearning.AI
- Certificate of Completion Claude Code in Action— by Anthropic
- MCP Course Unit 3: Automation in Production— by Hugging Face
- Certificate of completion: Claude 101— by Anthropic
- Certificate of completion: Introduction to Claude Cowork— by Anthropic

LANGUAGES

Mother Tongue(s): Oriya, Hindi

Dutch: Beginner Ready to learn if selected

English: Listening B2 | Reading B2 | Writing B2 | Spoken Production C1 | Spoken Interaction C1

EDUCATION

Master of Philosophy (M.Phil) in Commerce

2017

P.G. Department of Commerce, Utkal University, Bhubaneswar

Master of Commerce (M. Com)

2015

P.G. Department of Commerce, Utkal University, Bhubaneswar | With Concentration in Research Preparation